

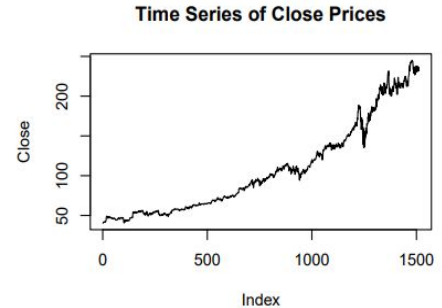
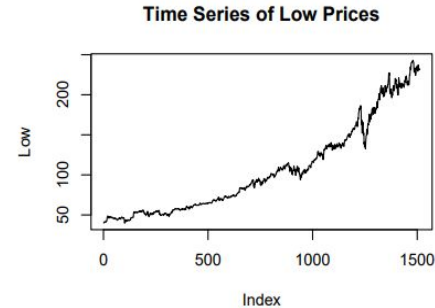
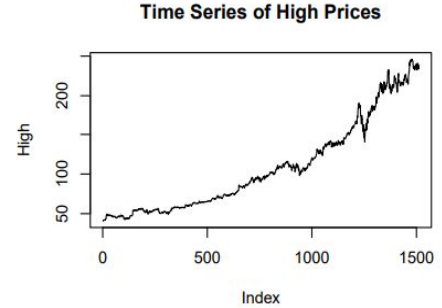
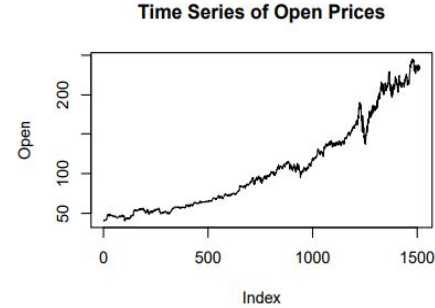
Microsoft Stock Data Analysis

Introduction

- This report analyzes Microsoft's stock data, covering variables like Open, High, Low, Close and Volume.
- Summary statistics : Mean, Median, Standard deviation.
- Key objectives :
 - - Identify patterns, trends and randomness in stock prices
 - - Use exploratory data analysis and statistical tests
- (Analysis of stock data, including exploratory data analysis, statistical tests, and non-parametric kernel regression.)

Exploratory Data Analysis

- Time Series plots of Microsoft's stock prices (Open, High, Low, Close) and Volume reveal significant trends.
- Observed upward trend across most variables.

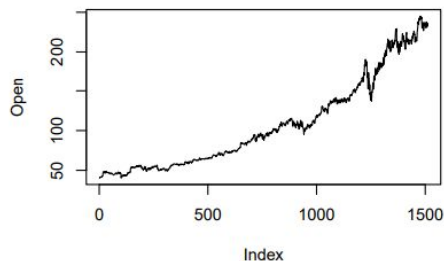


Summary Statistics and Time Series Plots

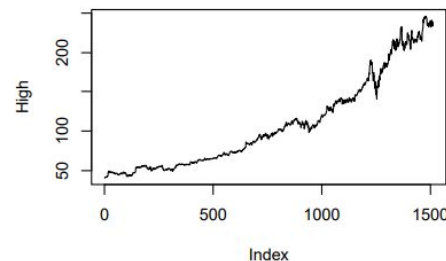
Open		High		Low	
Min.	: 40.34	Min.	: 40.74	Min.	: 39.72
1st Qu.:	57.86	1st Qu.:	58.06	1st Qu.:	57.42
Median :	93.99	Median :	95.10	Median :	92.92
Mean :	107.39	Mean :	108.44	Mean :	106.29
3rd Qu.:	139.44	3rd Qu.:	140.32	3rd Qu.:	137.82
Max.	:245.03	Max.	:246.13	Max.	:242.92

Close		Volume	
Min.	: 40.29	Min.	: 101612
1st Qu.:	57.85	1st Qu.:	21362129
Median :	93.86	Median :	26629615
Mean :	107.42	Mean :	30198625
3rd Qu.:	138.97	3rd Qu.:	34319616
Max.	:244.99	Max.	:135227059

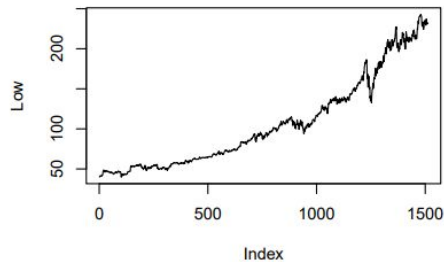
Time Series of Open Prices



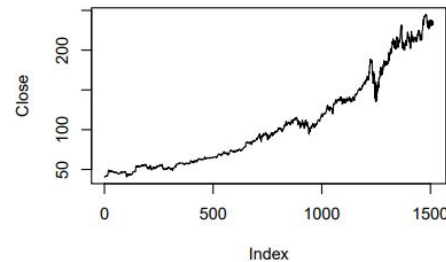
Time Series of High Prices



Time Series of Low Prices



Time Series of Close Prices



Run Test for Randomness

- The run test assesses randomness in stock prices :
- Result : Non-random clustering over time in prices and volume
- Volume shows higher values in earlier periods, indicating inverse relationship with price trends.

Mann-Kendall Trend Test

- The Mann-Kendall test detects monotonic trends in the time series data :
- - Significant upward trends in price variables (Open, High, Low, Close)
- - Suggests a consistent rise in stock prices over time.

Run Test and Mann-Kendall Trend test

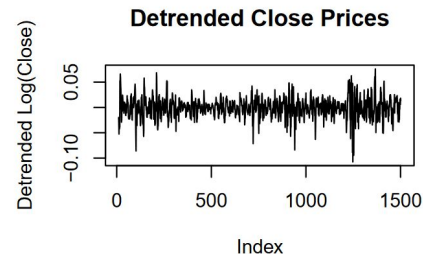
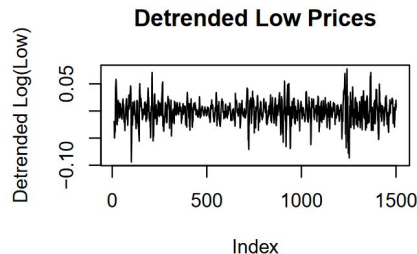
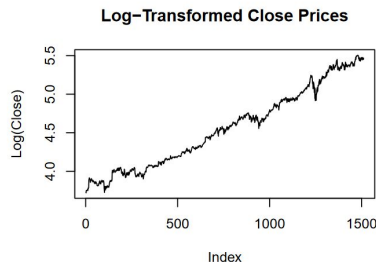
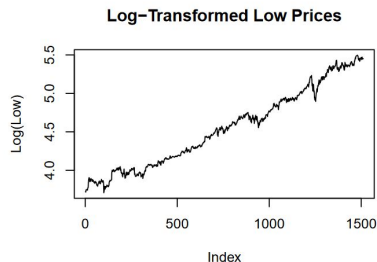
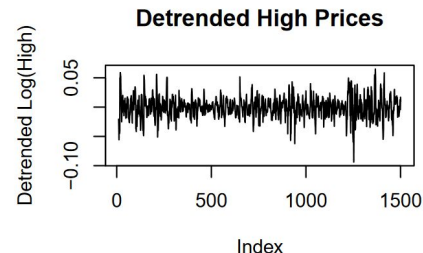
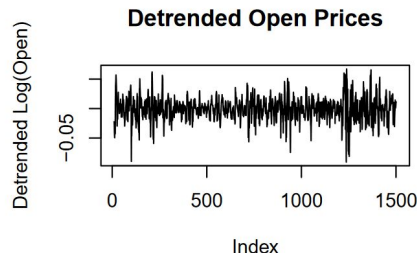
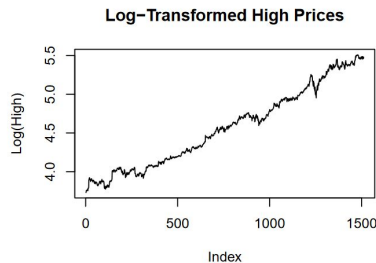
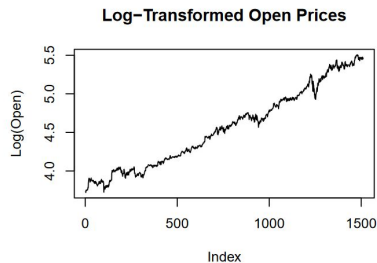
- Both Run test and Mann-Kendall trend test suggests significant monotonic upward Trend.
- To further analyze the data, we detrend each log-transformed variable by subtracting a 20-period moving average. This allows us to remove any remaining trend in the data.

Log Transformation and Detrended Analysis

To reduce the exponential trends:

- Log Transformation applied to variables.
- Detrending with a moving average technique.
- Increased randomness post-detrend, allowing for further statistical tests.

Log Transformation and Detrended Analysis



Run Test on Detrended Data

- After detrending the data, higher p-value in Run test Suggests that the data now behaves more randomly.
- We still get low run but data seems to be more random. After removing the trend the run value is increased.
- We can now use non-parametric tests for our analysis, since sequence is random.

```
## Detrended Close Run Test on Detrended Data  
## Test Statistic: -19.2161  
## p-value: 2.714362e-82
```

```
## Detrended Open Run Test on Detrended Data  
## Test Statistic: -20.14841  
## p-value: 2.778728e-90
```

```
## Detrended High Run Test on Detrended Data  
## Test Statistic: -21.28791  
## p-value: 1.469257e-100
```

```
## Detrended Low Run Test on Detrended Data  
## Test Statistic: -22.94537  
## p-value: 1.639108e-116
```

```
## Detrended Volume Run Test on Detrended Data  
## Test Statistic: -10.56626  
## p-value: 4.271766e-26
```

Two-Sample Tests for Location and Scale

- Location and Scale comparisons were conducted:
- Scale Alternative - Ansari-Bradley Test
- Location Alternative - Wilcoxon Rank Sum Test
- Higher p-values indicate similar central tendencies and variability among variables.

Scale Alternative Test (Ansari-Bradley Test)

- Null Hypothesis: Scale parameter(θ) = 1
- **Close vs. Open:** The p-value (0.94) suggests that the variability between Close and Open prices is similar.
- **Close vs. High and Low:** Close prices show similar variability compared to both High ($p = 0.92$) and Low ($p = 0.94$).
- **Open vs. High and Low:** Open prices also have similar variability to both High ($p = 0.97$) and Low ($p = 0.87$).
- **High vs. Low:** The variability between High and Low prices is not significantly different ($p = 0.86$), indicating similar scale.

Location Alternative Test (Wilcoxon Rank Sum Test)

- Null Hypothesis: The two independent samples come from identical populations with equal medians.
- **Close vs. Open:** With a p-value of 0.99, the test shows no significant difference in the median between Close and Open prices.
- **Close vs. High and Low:** Close prices have similar medians to both High ($p = 0.52$) and Low ($p = 0.47$) prices.
- **Open vs. High and Low:** Open prices also show no significant difference in median compared to both High ($p = 0.51$) and Low ($p = 0.48$).
- **High vs. Low:** The median for High and Low prices is similar, as indicated by a p-value of 0.20.

Two-Sample Run Test for Distribution Comparison

- Null Hypothesis: feature_1 distribution = feature_2 distribution
- **Close vs. Open:** The test shows Close and Open prices have very similar distributions after removing trends ($p = 1.00$).
- **Close vs. High and Low:** Close prices are also very similar to both High ($p = 0.93$) and Low ($p = 0.89$) prices in distribution.
- **Open vs. High and Low:** Open prices are similar in distribution to both High ($p = 0.86$) and Low ($p = 0.84$) prices.
- **High vs. Low:** High and Low prices show the most difference, but still aren't significantly different in distribution ($p = 0.40$).

Spearman Rank Correlation

- Null Hypothesis: Feature 1 and Feature 2 are not Independent.
- Spearman rank correlation measures monotonic relationships :
 - - High correlations between price variables (Open, High, Low, Close)
 - - Low correlations between Volume and price variables, suggesting weaker relationships

Results from Spearman Rank Correlation test

Variables	Correlation_Coefficient	p_value
Close & Open	0.999404	0.0000000
Close & High	0.999678	0.0000000
Close & Low	0.999752	0.0000000
Close & Volume	0.006615	0.7972423
Open & High	0.999752	0.0000000
Open & Low	0.999647	0.0000000
Open & Volume	0.010142	0.6936468
High & Low	0.999621	0.0000000
High & Volume	0.014822	0.5648038
Low & Volume	0.001158	0.9641175

Complimentary Analysis on Cats and Dogs Dataset

- Additional non-parametric analysis on audio features from cats and dogs :
- - Compared features like Pitch, Spectral Centroid, and Zero Crossing Rate
- - Results show similar distribution characteristics between both datasets

Conclusion & Future Work

- Key insights and potential applications :
 - - Run test recommended for model evaluation in machine learning (can be used as stopping criteria)
 - - Extension of non- parametric methods to other complex data types (images, text)
 - - Non-parametric tests offer flexibility for varied data distributions and fields